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Designing for Time-Pressured Digital Situations

The Impact of Microinteractions in Online Ticket Booking for Users with ADHD

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Abstract

Digital self-service systems have become central to modern society, yet time-pressured tasks such as online ticket booking often impose high cognitive demands on users. For individuals with Attention Deficit Hyperactivity Disorder (ADHD), who may experience challenges related to executive functions and working memory, these environments can be particularly demanding and stressful. This thesis investigates how the design of microinteractions, small functional elements such as visual feedback, animations, and guiding cues, influences task performance and user experience for neurodivergent users under time pressure.

Using an experimental within-subject design, 10 participants with a formal ADHD diagnosis completed booking tasks using two interactive prototypes. Version A included enhanced microinteractions, such as real-time feedback, subtle animations, and a guiding mascot, while Version B functioned as a standard control interface without microinteraction support. Data was collected through objective performance metrics, specifically task completion time and error rates, as well as subjective measures of perceived stress, cognitive load, and usability using 7-point Likert-scale questionnaires.

The results show that Version A led to faster task completion times and a lower number of errors compared to the control condition. Furthermore, participants reported lower levels of perceived stress and cognitive load, alongside higher usability ratings when interacting with the enhanced interface. These findings suggest that microinteractions function as external cognitive support mechanisms that reduce extraneous cognitive load and improve both efficiency and user experience in time-pressured digital environments.

This study contributes to the field of Human-Computer Interaction (HCI) by identifying how specific interface design elements can support neurodivergent users in commercial, time-sensitive contexts. The findings highlight the importance of moving beyond neurotypical user assumptions and emphasize the role of microinteractions in creating more inclusive and accessible digital systems.

Keywords: ADHD, Microinteractions, Cognitive Load, User Experience (UX), Time Pressure, Digital Self-Service, Inclusive Design, Human-Computer Interaction.

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1. Introduction

Today an increasing amount of every-day tasks can be carried out through digital platforms, including time-sensitive tasks. One task that can be stressful is buying tickets to events online under time pressure. In these situations users must navigate through potentially complex user interfaces in a limited amount of time. The interface design plays a crucial role for the user experience and their experienced stress (Norman, 2013; Paas & van Merrienboer, 2020). This raises important questions about how different user groups, including individuals with Attention Deficit Hyperactivity Disorder (ADHD), experience time-pressured digital environments (Barkley, 1997; Kofler et al., 2011). This chapter introduces the background and motivation for the study, outlines the problem statement concerning microinteraction design in such contexts, and presents the purpose, research questions, scope, and structure of the thesis.

As digital services continue to replace traditional forms of interaction, users are increasingly required to complete tasks independently without direct assistance. This places more responsibility on the design of the interface to guide the user and support effective task completion. In time-pressured situations, such as ticket booking, this becomes especially important, as users must process information quickly while avoiding mistakes that may affect the outcome of the task. Even small design elements can influence how easily users understand the system and how confident they feel when interacting with it (Norman, 2013).

In this context, the concept of microinteractions becomes relevant, as these small interaction details can support users by providing feedback, preventing errors, and guiding them through different steps in the process (Saffer, 2013). While these elements may seem minor, they can play an important role in reducing uncertainty and helping users maintain focus during stressful situations. This is especially relevant in interfaces where time pressure and high demand increase the cognitive load on the user (Paas & van Merrienboer, 2020).

At the same time, not all users experience digital environments in the same way. Individuals with ADHD may face additional challenges when interacting with time-pressured systems, particularly due to difficulties related to attention, working memory, and executive functioning (Barkley, 1997; Kofler et al., 2011). This means that design choices that may seem neutral for some users can have a larger impact on others. Despite this, many digital platforms are still designed with a general user in mind, without fully addressing the needs of users with different cognitive profiles.

These aspects highlight the importance of further investigating how interface design, and specifically microinteraction design, can support users in time-pressured digital situations. By focusing on both performance and user experience, this study aims to contribute to a better understanding of how digital systems can be designed to be more accessible and supportive for a wider range of users.

1.1. Problem statement

Digital self service systems have become a central part of how individuals interact with organizations within transportation, banking, healthcare and the event sector. Particularly, booking systems for tickets are characterized by limitations in time and accessibility where the user must quickly make decisions and perform correct information entry.

Research shows that time-limitation affects decision-making, risk assessment, and performance (Maule & Svenson, 1993; Kocher & Sutter, 2006). In digital environments, time pressure can also increase cognitive load, thereby impairing both accuracy and efficiency. (Paas and van Merrienboer, 2020) In this context, interface design becomes crucial. Small design decisions, such as how feedback is given, how error messages are presented, or how progression is visualized, can affect the user's experience of control, stress, and cognitive load. These details are often described as microinteractions, small discrete interaction moments that together shape the user experience. Despite this, research on how microinteractions affect performance under time pressure is still limited, especially in commercial booking environments.

Attention Deficit Hyperactivity Disorder (ADHD) is a neuropsychiatric disorder that affects executive functions such as working memory, inhibitory control, and attentional regulation (Barkley,m 1997; Faraone et al., 2021). International consensus research shows that ADHD is associated with difficulties in managing complex tasks, maintaining focus, and regulating behaviour under cognitive load (Faraone et al., 2021). Working memory limitations are particularly relevant in situations that require simultaneous information processing and rapid response (Kofler et al., 2011). According to Cognitive Load Theory (Sweller, 1988), human working memory has limited capacity. When unnecessary or design related cognitive load increases, performance deteriorates. For people with ADHD, this load can be particularly problematic, as executive functions are already affected (Barkley, 1997). Despite this,

many digital systems are based on a neurotypical user model. Neurodiversity has received increased attention in HCI, but empirical studies investigating how specific design components affect users with ADHD are still relatively few.

Taken together, three gaps can be identified in current research. First, most ADHD-related HCI studies have focused on educational or assistive contexts rather than commercial, time pressured digital environments. Second, few studies isolate microinteractions as experimental variables to assess their direct effects. Thirds, there is a lack of research combining objective performance metrics with subjective measures of stress and cognitive load, especially focused on people with ADHD. These gaps highlight the need for experimental research investigating how microinteractions design influences both task performance and user experience under time pressure for neurodivergent users.

1.2. Purpose and research questions

Based on the problem statement, interface design plays a crucial role in user experience and users measured stress level interacting with digital platforms. People with Attention Deficit Hyperactivity Disorder (ADHD) can experience more difficulties concerning cognitive load and stress in stressful situations. Still there is a lack of valid studies investigating how certain design aspects as microinteractions affect these users. The purpose of this study is to:

“To investigate how different microinteraction designs affect task performance, error rates, and perceived stress during time-pressured online ticket booking for users with ADHD.”

To achieve this, two research questions have been defined. Each question covers a distinct area of the study, focusing on task performance and user experiences.

The first research question focuses on identifying how microinteraction design affects users performance based on error rate and implemented time for users with ADHD under time pressure. Before any conclusions can be drawn it is crucial to collect data of participants task completion time and error rate. According to this, the first research question is:

[1] How do different microinteraction designs affect task completion time and error rate for users with ADHD under time pressure

The second research question focuses on the user experience and how microinteraction design affects users perceived stress, cognitive load and usability during stress-sensitive tasks such as ticket booking to give insight into how the microinteraction design affects the subjective experience. Therefore the second research question is:

[2] How do different microinteraction designs influence perceived stress, cognitive load, and usability during time-pressured booking tasks for users with ADHD?

1.3. Scope and limitations

The study is limited to how microinteractions in a digital ticket booking system affect users performance, perceived stress and cognitive load under time pressure for users with ADHD. The aim is to experimentally analyze how variations in microinteractions, such as feedback, error handling and progression visualization, affect task time error rate and subjective experience, using a controlled interactive prototype rather than a fully implemented system.

This study has several limitations. First, the experiment is conducted in a controlled environment with simulated booking tasks, which may differ from real world situations where factors such as network problems, unexpected errors or disruptive surroundings can affect user performance and experience. Second, the study focuses only on microinteractions related to feedback, error prevention and progression. Other aspects such as layout, color scheme or typography, are not systematically investigated. Third, the sample is limited to adult participants with ADHD, which may affect generalizability to other age groups or users with other neurodiverse profiles. Fourth, the prototype used in the study is not linked to real bookings or payments, such as the users own private bank information. This means that participants do not experience the same actual consequences or stakes as in a real booking situation. As a result, the perceived stress during the task may be lower than what would occur in real time pressured bookings , which amy affect the study's results regarding stress levels and experience. Finally, the study aims to generate insights that can inform the design of more accessible and supportive digital interfaces, but the results are context specific and cannot be automatically generalized to all types of digital self-service systems or industries. These limitations clarify the study's framework and contribute to transparency about the conditions within which the conclusions can be interpreted.

Another limitation concerns the potential influence of ADHD medication on participants performance and their subjective experience. Since medication intake was not controlled across participants, and information about medication status was collected through optional answers, variations in attention, cognitive load, and stress levels may have been affected by whether participants were medicated during the test.

1.4. Disposition

This thesis is structured into six main chapters.

Chapter 1 introduces the background, problem statement, purpose, research questions, and scope of the study. It also outlines the motivation for investigating microinteraction design in time-pressured digital environments for users with ADHD.

Chapter 2 presents the theoretical framework, including previous research on microinteractions, cognitive load, ADHD, and decision-making under time pressure. This chapter provides the conceptual foundation for the study.

Chapter 3 describes the methodology and implementation, including the research design, data collection process, experimental setup, and methods of analysis. It also discusses validity, reliability, and ethical considerations.

Chapter 4 presents the results of the study, including both objective performance data and subjective user experience measures.

Chapter 5 discusses the findings in relation to the research questions and theoretical framework, and includes a critical reflection on the chosen method.

Finally, Chapter 6 concludes the study by summarizing the key findings, discussing implications, and suggesting directions for future research.

2. Theoretical framework

2.1 Microinteractions in digital interfaces

As previously stated, Microinteractions are small, discrete interactions in digital systems that provide the user with feedback about actions, visualize progression, or prevent errors (Paas and van Merrienboer, 2020). Saffer (2013, p. 2) defines these as "contained product moments that revolve around a single main task," often described as the functional "DNA" of a user interface (Saffer, 2013, p. 4). According to Saffer's framework, a microinteraction is composed of four essential parts: a trigger that initiates the action, rules that determine how the system responds, feedback that communicates the response to the user, and loops and modes that define the interaction's duration and repetition (Saffer, 2013, pp. 9–20). These include, for example, real-time validation of inputs, animations that show an action being registered, and visual indicators that guide the user through a process (Norman, 2013; Saffer, 2013). As illustrated in Figure 1, these components form a structured process that guides the user through a single functional moment.

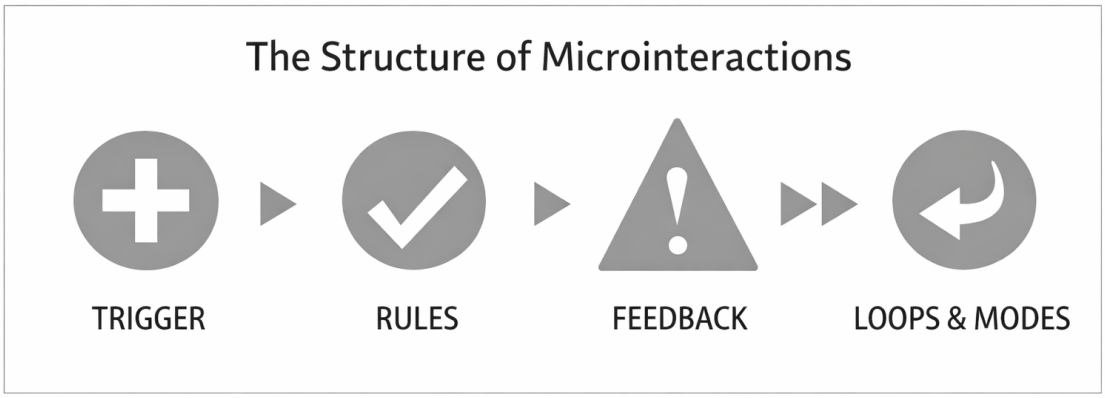


FIGURE 1. THE STRUCTURE OF A MICROINTERACTION. ADAPTED FROM MICROINTERACTIONS: DESIGNING WITH DETAILS, BY D. SAFFER, 2013, O'REILLY MEDIA.

In this framework, the trigger acts as the starting point, followed by the rules which govern the system's logic. The feedback then illuminates these rules to the user, while loops and modes determine how the interaction persists or changes over time (Saffer, 2013).

The communication between the system and the user is crucial for the overall experience. Norman (2013, p. 25) emphasizes that clear and immediate feedback is necessary for the user to understand the system's status and to avoid uncertainty about whether their action has been registered (Norman, 2013, pp. 38–40). Beyond simple feedback, microinteractions utilize what Gaver (1991, pp. 79–80) describes as perceptual affordances. This means that the design itself provides visual cues on how

it should be used, allowing the user to navigate without having to guess or read complex instructions.

Regarding animations, research by Vogel and Balakrishnan (2005, pp. 137–138) suggests that subtle visual cues help the user maintain focus and create a sense of continuity. This prevents the user from becoming confused or distracted when the interface changes quickly. Research in HCI shows that such well-designed microinteractions can improve both performance and experience by reducing uncertainty and making it easier to navigate complex digital environments (Bargas-Avila & Hornbæk, 2011).

In the context of time-sensitive tasks, such as online ticket booking, microinteractions can help users understand the system more quickly, avoid errors, and feel in control of the process. By providing constant and subtle guidance, they act as external supports that simplify decision-making and reduce the load on working memory (Vogel & Balakrishnan, 2005). Despite the recognized importance of microinteractions, there is limited empirical research that systematically examines their effect on performance and experience under time pressure, especially for users with ADHD. Studies that isolate microinteractions as experimental variables are few, creating a knowledge gap that this study addresses.

2.2 Cognitive Load and Working Memory

Cognitive Load Theory (Sweller, 1988; Paas & van Merriënboer, 2020) describes how our working memory has a limited capacity. This means we can only process a certain amount of information at once before it becomes too much and we start making mistakes (Miller, 1956). Within this theory, three different types of load are identified:

- **Intrinsic load:** The complexity of the task itself, such as how many steps a ticket booking actually requires.
- **Extraneous load:** Unnecessary or poorly designed information that loads working memory. This is where microinteractions come in to "clear up the clutter."
- **Germane load:** the cognitive effort that actually contributes to learning or problem solving (Sweller, 1988).

In digital interfaces, microinteractions work as a tool to reduce the unnecessary "extraneous load" by guiding the user and providing direct feedback (Paas & van Merriënboer, 2020). According to Mayer and Moreno (2003), it is crucial to remove

"noise" in the design when a task is mentally demanding. Microinteractions act as a form of mental support by giving the user the right cue at the exact right moment, so they don’t have to waste energy guessing what happened (Mayer & Moreno, 2003).

This becomes especially important in situations with time pressure. When we are stressed, the resources in our working memory are even more restricted. Having a system that supports the user through small, clear steps is therefore critical for keeping stress levels down and ensuring that the booking is done correctly (Mayer & Moreno, 2003; Paas & van Merriënboer, 2020).

2.3 ADHD and Executive Functions

Studies show that individuals with ADHD often have difficulty planning, organizing and managing complex tasks, especially when multiple sources of information need to be simultaneously (Kofler et al., 2011). According to Barkley (1997), ADHD is closely linked to challenges with executive functions, which include working memory and the ability to regulate one's attention and impulses. This means that navigating a complex interface isn't just a matter of following steps, it is a constant drain on the user's mental energy.

In digital environments, this often leads to users with ADHD experiencing higher cognitive load and stress during time-pressured tasks compared to neurotypical users. When a system is slow to respond or lacks clear feedback, it creates a "temporal gap" that can make it harder for someone with ADHD to stay focused on the task (Antrop et al., 2005). This is why supporting mechanisms like microinteractions are so important. By providing constant "external cues" and immediate feedback, microinteractions help bridge these gaps, reducing the load on working memory and making the booking process much more manageable (Antrop et al., 2005; Barkley, 1997).

2.4 Combining objective and subjective measures

Previous research in HCI and ADHD has often focused on either objective measures (e.g. time, errors) or subjective experiences (e.g. stress, usability). Bargas- Avila and Hornbeak (2011) point out the importance of combining these measures to get a holistic picture of the users performance and experience. By measuring both task completion time, error rate and subjective experiences of stress and cognitive load,

this study can analyze how microinteractions affect both performance and experience.

2.5 Decision making under time pressure

Time pressure has shown to significantly influence how individuals process information and make decisions. Under constrained time conditions, individuals tend to rely on simplified decision strategies, often leading to a trade off between speed and accuracy (Payne et al., 1996;Kocher and Stutter, 2006). Time pressure has been found to affect how individuals evaluate risk and probabilities. For example, Diana L. Young et al. (2012) show that time pressure can increase the attractiveness of risk in decision making situations involving potential gains. Their findings suggest that individuals under time pressure are more likely to engage in risk seeking behaviour, even when no changes occur in outcome utility or probability evaluation (Young er al., 2012, p. 180-181).

Furthermore, time pressure can impair individuals ability to discriminate between probabilities, particularly in loss contexts. This reduces probability sensitivity means that users may struggle accurately assess differences between options, potentially leading to suboptimal decisions (Young et al., p 187). These findings indicate that time pressure does not only affect the speed of decision making but also the underlying cognitive mechanisms that guide user behavior.

Wu et al. (2022) studied how time pressure influences the way people explore options and make decisions. They used a version of the *four-armed bandit task,* where participants experienced different levels of expected reward, uncertainty, and time pressure (limited vs. unlimited time). The study found that time pressure reduced exploration based on uncertainty and made participants repeat their previous choices more often. Participants also became less sensitive to differences in rewards, leading to lower average rewards under time pressure compared to when they had unlimited time (Wu et al., 2022).

Looking at reaction times and how participants gathered information, the researchers found that options with higher expected rewards are chosen more quickly, while

uncertainty on how long decisions took was smaller, showing that people tended to use simpler, less mentally demanding strategies (Wu et al., 2022). The results suggest that when people are under time pressure, they use resource efficient strategies and focus on familiar options and repeat past choices instead of exploring uncertain alternatives. This helps us understand how cognitive limits influence exploration and decision making (Wu et al., 2022).

3. Method and implementation

This chapter explains the method and implementation of the study. It describes the recruitment of participants, how data will be collected and how the data will be analysed. The purpose of this part is to explain how the study was conducted to answer the research questions.

The study is an experimental study with a mixed-methods approach, combining quantitative measures of task performance with qualitative estimations of user experience. This design allows us to study how microinteraction designs affect task performance, perceived stress, and usability during a time-pressured online ticket booking task for users with ADHD.

The choice of an experimental mixed-methods design is motivated by the need to capture both objective performance and subjective user experience. In Human-Computer Interaction research, combining quantitative and qualitative methods is recommended to provide a more complete understanding of how users interact with digital systems (Bargas-Avila & Hornbæk, 2011). Mixed methods approaches are widely used in HCI research because they allow researchers to combine different types of data. For example, researchers can collect measurable data about how users behave, while also gathering information about hoe users experience a system or service. According to Creswell and Plano Clark (2018), using both quantitative and qualitative methods together makes research findings more reliable. This is because the two approaches complement each other and provide a more complete understanding of the same phenomenon. This is especially important in usability studies, where both how well something performs and how users perceive it are equally important.

An experimental approach was chosen to allow for controlled comparison between two interface conditions, making it possible to isolate the effect of microinteraction

design on user performance and experience. By keeping the prototypes identical except for the microinteractions, differences in results can be attributed to the design variation (Norman, 2013).

Furthermore, using a within-subject design, meaning that each participant interacted with both prototypes, reduces the impact of individual differences. This is particularly important when studying users with ADHD, as cognitive abilities may vary between individuals (Barkley, 1997).

3.1. Data collection

A total of 10 participants were recruited with a formal ADHD diagnosis. Inclusion criteria for the ADHD group included a formal diagnosis and age between 18 and 50. All participants provided informed consent prior to participation.

The experiment involved two prototypes of an online ticket booking platform, identical in design and content but differing in microinteraction implementation:

- Version A (experimental): includes microinteractions such as subtle animations, visual feedback, and a mascot providing nonverbal guidance.
- Version B (control): standard interface without microinteractions.

The booking task consisted of five main steps: homepage, date selection, seat selection, payment, and confirmation. Each participant completed the task twice, once on each prototype. To reduce learning effects, half of the participants started with Version A and half with Version B.

Objective performance measures included:

- Task completion time, recorded automatically by a timer built into the prototype.
- Error rate, recorded automatically, including mistakes such as incorrect email or credit card entries. Participants used a test card to ensure realistic input without compromising privacy.

After each task, participants completed a questionnaire measuring perceived stress, cognitive load, usability, and overall user experience.

This approach ensured the collection of both objective performance data and subjective experience data, enabling direct comparisons between interface versions and participant groups.

3.2. Data analysis

This section describes the methods used to analyze the collected data, connecting them to the research questions and the data collection procedures.

For quantitative data, including task completion time and error rate, the information will get collected and calculated for each participant and prototype. Collected data will be analyzed and prepared between Prototype A and B to decide whether microinteraction design significantly affects task performance. The results will be visualized in tables to make comparisons clear and visualised.

To analyze differences between Prototype A and Prototype B, a Wilcoxon Signed-Rank Test was performed. This non-parametric test was chosen because the study used a within subject design and included a relatively small sample size (n = 10). The test was applied to task completion time and error rate to determine whether observed differences between the prototypes were statistically significant.

For qualitative data from questionnaires about perceived stress, cognitive load, usability, and overall user experience, open-ended responses will be analyzed thematically to identify common experiences and patterns.

This mixed-methods approach ensures that both objective performance and subjective experience are captured, allowing comprehensive analysis of how microinteraction designs impact users under time pressure.

3.3. Validity and reliability

The prototypes were designed with the exact same interface in terms of layout, structure, content, and navigation. To ensure that the design would not make an impact on the results the prototypes were identical with use of images, colour and all design aspects. The only difference between them was that prototype A included microinteractions such as subtle animations and guiding cues together with a mascot providing guidance and feedback during the booking process. This ensured that any observed differences in performance or user experience could be connected to the microinteraction design.

The booking task was designed to imitate a typical online ticket purchasing process. The interface was built based on several existing ticket website designs, ensuring that the layout would not impact the users results. The task consisted of six stages: homepage, category page, date selection, seat selection, payment, and confirmation. These steps were chosen because they reflect the standard flow of ticket booking systems. By including all these stages, the task provided a realistic scenario under time pressure and ensured that participants get the experience of a real ticket booking process.

The objective measurements that were used during the test were task completion time (time spent to finish the task) and error rate (registered errors during the task). Task completion could be measured by a built in timer in the prototype which gave an exact measure on the task completion time. Since the prototype was built in Figma the error rate could not be measured automatically, to avoid bias in affecting the participants stress level the test was screen recorded. This made it possible to gather the data after the participants finished the test.

To ensure that the measurements of performance were collected correctly, an "error" was defined as any variation from the optimal task path or specific instructions. In the analysis of the screen recordings, errors were categorized into three types:

- **Instructional Errors:** Selecting the wrong date or an incorrect number of seats.
- **Input Errors:** Entering an incorrect email format or leaving mandatory fields empty.
- **Process Errors:** Repetitive clicking on buttons due to uncertainty or perceived system lag.

This made it easier to carefully go through errors without having to observe them directly. In that way the test could get performed realistically, without unnecessary stress and the reliability in the data could be ensured.

After completing each booking task, participants filled out a questionnaire designed to measure their subjective experience. The questionnaire included:

- **Perceived stress** - how stressed participants felt during the task.
- **Cognitive load** - how mentally demanding they perceived the task to be.
- **Usability** - how easy and clear the interface was to use, including clarity of feedback and guidance provided by microinteractions such as the mascot.

The questionnaire was implemented as an online form (e.g., Google Forms) to allow participants to provide responses immediately after the task, ensuring that their responses were accurate to the task completed. Each participant completed the questionnaire twice, once after testing each prototype. The questionnaire was short and based on nine likert scale questions, ranging from “very low” to “very high,” followed by 3 open ended questions where the participants could provide more in-depth feedback.

The objective measures, task completion time and error rate, provide concrete data on how microinteraction design affects users’ performance under time pressure. This allows us to address RQ1 (“How do different microinteraction designs affect task completion time and error rate for users with ADHD under time pressure?”)

The subjective measures, perceived stress, cognitive load, and usability, provide insights into the participants’ experience and how microinteractions influence their stress level and cognitive load. These measures are directly connected to RQ2 (“How do different microinteraction designs influence perceived stress, cognitive load, and usability during time-pressured booking tasks for users with ADHD?”) These measures work together with the objective data to give a clear view of both how well participants performed and how they experienced the task.

By using both objective and subjective measures, we can look at both task performance and user experience, which helps us understand how microinteractions affect users with ADHD.

All participants got the same instructions and same tasks to perform with identical steps: homepage, category page, date selection, seat selection, payment, and confirmation. The task was designed to be the same for everyone, so that performance and experience could be compared across prototypes and participant groups. The only difference was the order of the prototypes: half of the participants started with Prototype A and half started with Prototype B, to balance possible learning effects.

The tests were conducted in the participants’ own homes using their personal computers. Participants were instructed to choose a quiet location free from distractions, such as other people in the room or background noise, in order to minimize external factors that could affect the results. Conducting the test at home on the participants’ own computers created a more realistic setting and reduced the potential influence of using an unfamiliar device or environment. All participants used the same type of web browser, and access to a stable internet connection was ensured prior to the test to avoid technical issues. Since the experiment was carried out in a home environment, variations as screen size or lighting conditions could occur. These

were considered as real-world usage scenarios and reflected the natural user environment for online ticket booking.

3.4. Considerations

Ethical considerations were an important aspect of the design and implementation of this study. Since the research involves human participants and investigates experiences of stress in a time pressured digital environment, several ethical principles were followed to ensure participants' well being, privacy and autonomy.

Informed consent

Before participating in the study, all participants received written information describing the purpose of the research, the procedure of the experiment, and how the collected data would be used. The information also explained that participation was voluntary and that the study focused on interaction with a prototype ticket booking system under time pressure. After receiving this information, participants provided informed consent before taking part in the experiment.

Voluntary participation and the right to withdraw

Participation on the study was completely voluntary. Participants were informed that they had the right to withdraw from the study at any time without providing a reason and without any negative consequences. If a participant had chosen to withdraw, any collected data from that participant would have been removed from the dataset.

Protection of participants with ADHD

Because the study included participants with ADHD, special care was taken to avoid stigmatization or framing ADHD as a deficit. The research was presented as an investigation of how interface designs work for different types of users, rather than evaluating the abilities of individuals. Results are reported at a group level rather than at an individual level to prevent identification or labeling of specific participants.

Privacy and data protection

The study did not collect sensitive personal information such as names, addresses, or personal identification numbers. All collected data were anonymized and stored without any identifiable information. Screen recording used to identify errors during the task were reviewed only for research purposes and were not shared outside research process.

Medication considerations

Since the study includes participants with ADHD, the potential influence of medication on cognitive performance and stress was taken into consideration. Participants who reported having a prescribed ADHD medication were informed prior to the experiment that they could complete the test either before or after taking their medication. If possible, they were encouraged to complete the task before medication intake. However, this was entirely voluntary and no requirement was imposed.

To account for this factor in the analysis, participants were asked in the post task questionnaire whether they had been under the effect of ADHD medication during the test. This question was optional, allowing participants to choose not to disclose this information. This approach ensured that participants integrity and well being were respected, while still enabling the study to consider medication as a potential influencing factor on task performance, perceived stress, and cognitive load.

Use of test payment information

To simulate a realistic booking process without exploring the participants privacy a fake credit card was used instead of an authentic one. This way, the booking process is simulated without any personal or economical information being needed.

Managing stress during the experiment

Because the study explores perceptions during time pressure, there was a chans that the participants would experience stress during the task. To minimize unnecessary stress that could interfere with the authenticity of the experiment, participants fulfilled the tasks at home or were they felt comfortable and in a prototype environment, and not through a real booking system. The participants were also informed beforehand that the bookings are fictitious and that no real tickets or payments will go through. This decreased the risk of worry linked to economical consequences or mistakes during the task.

Societal relevance

The purpose of this study is to contribute to a more including and available digital design. Through researching how microinteractions affect users with various attention-profiles the results can contribute to the development of digital services that allow support for people with ADHD and other cognitive differences. This contributes to the broader goal of designing digital systems that are usable and supportive for a diverse range of users.

4. Results

This chapter presents the findings from the user tests conducted with 10 participants diagnosed with ADHD. The results are divided into two sections: first, the objective presentation of collected data, followed by the data analysis.

4.1 Presentation of Collected Data

The data presented in this section was collected through prototype logging, screen recordings, and post-task questionnaires. The results are presented objectively without interpretation.

4.1.1 Task Completion Time and Error Rate

An error was defined as any deviation from the intended task flow or instructions. Errors were categorized into three types: instructional errors, input errors, and process errors.

Table 1 shows the individual results for task completion time (TCT) and total error counts for both prototypes.

The mean task completion time for **Prototype A** was **219 seconds**, compared to **250 seconds** for **Prototype B**.

The mean number of errors for **Prototype A** was **1.7**, while **Prototype B** resulted in a higher mean of **4.1 errors**.

Across participants, most completed the task faster and with fewer errors in Prototype A. However, some variation was observed. For example, participants P3 and P8 had shorter completion times in Prototype B, but still showed differences in error rates.

TABLE 1 INDIVIDUAL TASK PERFORMANCE METRICS FOR PROTOTYPE A AND B

Participant	Start Order	Time A (s)	Total Errors A	Time B (s)	Total Errors B
P1	A → B	210	1	265	5
P2	B → A	195	1	240	3
P3	**A → B**	**255**	**4**	**215**	**2**
P4	B → A	190	1	280	6
P5	A → B	225	2	245	3
P6	B → A	205	1	225	2
P7	A → B	195	1	290	8
P8	**B → A**	**270**	**3**	**225**	**2**
P9	A → B	215	1	255	5
P10	B → A	230	2	260	5
Mean		**219**	**1.7**	**250**	**4.1**

THIS TABLE PRESENTS INDIVIDUAL AND MEAN SCORES FOR TASK COMPLETION TIME (TCT) AND ERROR COUNTS. BOLDED ROWS HIGHLIGHT PARTICIPANTS (P3 AND P8) WHO DEVIATED FROM THE GENERAL TREND.

4.1.2 Self-Reported Stress, Cognitive Load, and Usability

Subjective data was collected using a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). Table 2 presents the mean values for each measured construct.

For perceived stress, participants reported lower stress levels in Prototype A. The statement *"I felt a high level of stress during the task"* had a mean of **3.4** for Prototype A and **5.8** for Prototype B. Similarly, *"Time pressure made it difficult to focus"* scored **3.6** for Prototype A and **6.2** for Prototype B.

For cognitive load, the task was rated as less mentally demanding in Prototype A (**3.8**) compared to Prototype B (**5.9**). The interface was also rated as easier to understand in Prototype A (**5.8**) than in Prototype B (**4.1**).

For system usability, Prototype A received higher ratings across all measures. Participants reported a higher sense of control (**5.2 vs. 2.4**), clearer feedback (**6.4 vs. 1.9**), stronger confidence in their actions (**6.3 vs. 2.2**), and better support from visual cues (**6.1 vs. 2.1**).

The guidance element (mascot) in Prototype A received a mean score of **5.1**.

TABLE 2 MEAN RATINGS FOR SUBJECTIVE USER EXPERIENCE AND COGNITIVE LOAD CONSTRUCTS (N = 10)

Construct	Statement / Metric	Prototype A	Prototype B
Perceived Stress	I felt a high level of stress during the task	3.4	5.8
	Time pressure made it difficult to focus	3.6	6.2
Cognitive Load	I found the task to be mentally demanding	3.8	5.9
	The interface layout was easy to understand	5.8	4.1
System Usability	I felt in control of the booking process	5.2	2.4
	The system provided clear and helpful feedback	6.4	1.9
	I felt confident that my actions were registered	6.3	2.2
	Visual cues helped me navigate the steps	6.1	2.1
Guidance	The assistant (mascot) supported my focus	5.1	N/A

ALL ITEMS WERE RATED ON A 7-POINT LIKERT SCALE (1 = STRONGLY DISAGREE, 7 = STRONGLY AGREE). THE N/A VALUE INDICATING "NOT APPLICABLE" REFERS TO THE ABSENCE OF THE MASCOT IN PROTOTYPE B.

4.1.3 Qualitative Feedback

Participants answered open-ended questions after completing the tasks. Their responses were grouped into three themes.

Theme 1: Stress in the Booking Process

Participants described different sources of stress between the prototypes. In Prototype

A, most participants reported low stress levels. However, a few participants (P3 and P8) mentioned that animations could be distracting.

In Prototype B, most participants described uncertainty as a main source of stress. Several participants mentioned that they were unsure whether their actions had been registered.

Theme 2: Elements Supporting Focus

Participants highlighted different elements that helped them stay focused.

In Prototype A, the progress bar was frequently described as helpful for understanding the process. The mascot was also mentioned as supportive by several participants.

In Prototype B, some participants noted that the lack of animations made it easier to focus on reading information.

Theme 3: Distractions and Confusion

Participants reported different types of confusion across the prototypes.

In Prototype B, participants described difficulties identifying errors due to lack of immediate feedback.

In Prototype A, a small number of participants described animations or moving elements as distracting.

4.2 Data Analysis

The results show clear differences between Prototype A and Prototype B in both task performance and user experience.

Task performance (RQ1)

Prototype A resulted in a lower mean task completion time (219 seconds) compared to Prototype B (250 seconds). In addition, participants made fewer errors in Prototype A (mean = 1.7) than in Prototype B (mean = 4.1).

Most participants performed better in Prototype A, both in terms of speed and accuracy. However, some variation was observed. For example, participants P3 and P8 completed the task faster in Prototype B, although this did not consistently result in fewer errors. This indicates that while the general trend favors Prototype A, individual differences exist.

User experience (RQ2)

The subjective results show consistent differences between the prototypes. Participants reported lower perceived stress and lower cognitive load in Prototype A compared to Prototype B.

Prototype A also received higher ratings in all usability-related measures, including sense of control, clarity of feedback, confidence in actions, and usefulness of visual cues. These results indicate that participants experienced Prototype A as easier to use and more supportive during the task.

Comparison of objective and subjective results

The objective and subjective data follow the same pattern. Prototype A is associated with both improved performance (faster time and fewer errors) and a more positive user experience (lower stress and higher usability ratings).

The qualitative responses support these findings. Participants described Prototype A as clearer and more supportive, while Prototype B was often associated with uncertainty and lack of feedback.

At the same time, the qualitative data shows that some participants experienced elements in Prototype A, such as animations, as distracting. This suggests that while microinteractions may support many users, they may not be equally beneficial for all individuals.

To evaluate whether the observed differences between the prototypes were statistically significant, Wilcoxon Signed-Rank Tests were conducted.

For task completion time, the analysis showed no statistically significant difference between Prototype A and Prototype B ($W = 11.0$, $p = .105$), despite the lower mean completion time observed for Prototype A.

For error rate, the analysis revealed a statistically significant difference between the prototypes ($W = 6.5$, $p = .037$). Participants made significantly fewer errors when using Prototype A compared to Prototype B.

These results suggest that the microinteraction-enhanced interface had a measurable positive effect on accuracy, while the improvement in task completion time could not be confirmed statistically within the present sample.

5. Discussion

This chapter discusses the findings of the study in relation to the research questions and the theoretical framework. The aim is to interpret the results and reflect on how microinteraction design influences both task performance and user experience in time-pressured digital environments for users with ADHD.

The discussion begins by revisiting the purpose of the study and the two research questions, focusing on how microinteraction design affects (1) objective performance measures, including task completion time and error rate, and (2) subjective user experience, including perceived stress, cognitive load, and usability. The results are analyzed in relation to existing research on cognitive load, ADHD, and interaction design.

Furthermore, this chapter explores possible explanations for the observed results, considering both the benefits and limitations of microinteraction design. Particular attention is given to how different design elements may support or hinder users depending on individual cognitive differences.

The chapter concludes with a discussion of methodological considerations, including strengths and limitations of the chosen research design, as well as implications for future research and design practice.

5.1. Result discussion

This study aimed to investigate how different microinteractions designs affect task performance, error rates, and perceived stress during time pressured online booking for users with ADHD. The two research questions focused on (1) objective performance outcomes and (2) subjective user experience.

Discussion of task performance (RQ1)

The result show how prototype A, that included microinteractions, lead to faster task completion times and fewer errors compared to Prototype B. This suggest that microinteractions can play a significant role in supporting users with ADHD in time pressured digital environments. One explanation to these results can be understood by cognitive Load Theory (Sweller, 1988; Paas and Van Merrienboer, 2020). Microinteractions in Prototype A, relating to feedback in real time, and progress indicators, likely reduced outside cognitive load by guiding the user through the task. This aligns with earlier research that indicate clear system feedback reduces uncertainty and improves task efficiency (Norman, 2013; Mayer and Moreno, 2003).

Furthermore the reduced error rate in Prototype A indicates that microinteractions not only improve speed but also support accuracy, which os particularly important unde time pressure. This also supports earlier findings about users under time pressure tend to rely on simplified decision making strategies (Payne et al., 1996; Wu et al., 2022). By providing immediate feedback and guidance, microinteractions may help users avoid these shortcuts and make more accurate decisions.

However, the variation observed in participants P3 and P8 suggest that microinteraction design is not universally beneficial. In some cases, participants performed faster in the non microinteraction design is not universally beneficial. In some cases, participants performed faster in the non microinteraction condition. This may indicate that certain design elements, such as animations, can introduce addition cognitive load for some users, particularly if they are perceived as distracting. This shows that microinteractions must be carefully designed to support rather then overwhelm users.

Although Prototype A resulted in faster average task completion times, the statistical analysis did not show a significant difference between the prototypes. This may be explained by the relatively small sample size, which reduces statistical power. A larger sample could potentially reveal a significant effect on task completion time.

In contrast, the reduction in error rate was statistically significant, suggesting that microinteractions primarily supported task accuracy rather than speed.

Discussion of user experience (RQ2)

The results strongly indicate that prototype A reduced perceived stress and cognitive load, while also improving usability and sense of control. These findings align with previous research showing that clear feedback and guidance can increase users confidence and reduce uncertainty in digital interfaces (Norman, 2013; Saffer, 2013).

From a ADHD perspective these findings are particularly important. Individuals with adhd often experience difficulties related to working memory and attention regulation (Barkley, 1997; Kofler et al., 2011). In time pressured situations, these challenged can be amplified. The microinteractions in prototype A may have functioned as external cognitive support, helping users maintain focus and stay oriented through out the booking process.

The qualitative results further support this interpretation. Participants described prototype A as clearer and more supportive, while Prototype B was associated with uncertainty and lack of feedback. This reflects what delayed or missing feedback

disrupts attention. Microinteractions appear to bridge these gaps, thereby improving both experience and performance.

At the same time, some participants reported that animations were distracting. This suggest that while microinteractions can reduce cognitive load for many users, they may also introduce sensory or attentional overload if not carefully balanced. This reinforces the importance of adaptable or customizable interface design.

Relation to previous research and theoretical framework

Overall, the findings of this study are consistent with existing research within Human Computer Interaction, particularly regarding the importance of feedback, affordances, and interaction clarity in digital interfaces (Norman, 2013; Gaver, 1991). The results support Norman's (2013) principles of design, which emphasize that clear and immediate feedback reduces uncertainty and helps users understand the system state. In this study, the microinteractions in Prototype A, such as real-time validation and visual progress indicators appear to have reduced the "gulf of evaluation," allowing users to more easily interpret whether their actions were successful. This likely contributed to both improved task performance and reduced perceived stress.

The findings can also be interpreted through Cognitive Load Theory (Sweller, 1988; Paas & van Merriënboer, 2020). Microinteractions seem to function as a mechanism for reducing extraneous cognitive load, which refers to unnecessary mental effort caused by poor design. By providing timely feedback and guiding users step by step, the interface in Prototype A minimized the need for users to internally process uncertainty or remember previous steps. This is particularly important in time pressured situations, where working memory resources are already limited. The results therefore support previous research suggesting that well designed interfaces can offload cognitive effort from the user and improve both efficiency and accuracy (Mayer & Moreno, 2003).

From an ADHD perspective, the results align with earlier research on executive functioning and working memory limitations (Barkley, 1997; Kofler et al., 2011). Individuals with ADHD often experience difficulties maintaining attention and managing multiple pieces of information simultaneously. In this study, microinteractions appear to have acted as external cognitive support by providing continuous cues and reducing the need for internal regulation. This supports the idea

that design can play a compensatory role, helping users manage cognitive challenges rather than placing the full burden on the individual.

At the same time, the findings also nuance previous research by showing that microinteractions are not universally beneficial. While most participants experienced reduced stress and improved performance, some reported that animations and moving elements were distracting. This suggests that certain types of microinteractions may introduce additional cognitive or perceptual load, particularly for users who are sensitive to visual stimuli. This finding extends Cognitive Load Theory by indicating that design elements intended to reduce extraneous load may, if poorly calibrated, instead contribute to it. It also highlights the importance of balance in interface design, where supportive elements should not become overwhelming.

Furthermore, the study contributes to the growing body of research on neurodiversity in HCI. Previous studies have often focused on educational or assistive technologies, whereas this study demonstrates that similar principles apply in commercial, time-pressured environments such as ticket booking systems. By combining objective performance measures with subjective user experience data, the study also responds to calls within HCI research for more holistic evaluation methods (Bargas-Avila & Hornbæk, 2011).

Finally, the results challenge the assumption of a "one-size-fits-all" user model in interface design. Rather than viewing difficulties as solely user related, the findings support a perspective where usability issues emerge from a mismatch between system design and user needs. In this sense, microinteractions can be understood not only as usability enhancements but as tools for inclusive design. By supporting attention, reducing uncertainty, and guiding user behavior, they contribute to the development of digital systems that are more accessible to users with diverse cognitive profiles.

Broader implications: designing beyond the "neurotypical user"

An important implication of these findings relates to how digital systems are typically designed. Many interfaces are implicitly built around a "neurotypical" user model, where assumptions are made about attention span, working memory, and ability to process information under pressure.

The results of this study suggest that such assumptions can unintentionally exclude users with different cognitive profiles, including individuals with ADHD. When systems lack clear feedback or require users to manage high cognitive load independently, they may become less accessible.

This perspective is supported by previous research indicating that individuals with ADHD often experience difficulties in adapting to environments that are not designed to accommodate their cognitive needs (Antrop et al., 2005). Rather than being solely a result of individual limitations, these challenges can be understood as emerging from a mismatch between user needs and system design. In this sense, digital environments may reinforce existing cognitive norms, where users who do not align with these norms experience increased difficulty in performing everyday tasks.

From this perspective, microinteractions can be seen not only as usability enhancements, but as tools for inclusive design. By reducing uncertainty, supporting attention, and guiding users through complex tasks, they can help create digital environments that are accessible to a wider range of users. This aligns with a broader shift in HCI towards designing for neurodiversity, where variability in cognitive functioning is considered a natural part of the user population rather than a deviation from a standard.

5.2. Method discussion

The purpose of this study was to explore how microinteraction design affects error rate, task performance and stress-level during time-pressured online ticket booking for users with ADHD. The chosen method for the experiment was mixed-methods and was a suitable method to answer both of the research questions.

By using a controlled experimental design with two prototypes that were identical in all aspects except for the microinteractions, the study was able to isolate the effect of microinteraction design. This strengthens the validity since differences in error rate, task competition time and user experience can be assigned to the use of microinteractions rather than other design factors.

Another strength is using a mixed-methods approach by combining quantitative data gathered by measuring task competition time and error rate together with qualitative data gathered by the questionnaire which collects data about the participants rating of experienced stress, cognitive load and open ended questions. This delivers a comprehensive understanding of the research problem which aligns well with the purpose of the study and allows both research questions to be answered. The quantitative data clearly answers RQ1 while the qualitative data addresses RQ2.

The study also benefits from a realistic task design with a booking process that includes typical steps that are found in real-world settings of a booking system. By

allowing participants to perform the test in their own environment at their own computer the test contributes to a more natural and realistic user experience.

There are still some limitations to the chosen method, the sample size was relatively small with 10 participants which limits the generalizability of the test results. Clear patterns were observed but a larger sample size would increase the reliability of the findings and allow for stronger statistical conclusions.

Further the task was designed to simulate a real booking process but it was still conducted in a prototype setting without real consequences. This may have reduced the level of stress experienced by participants, which affects the external validity, particularly regarding conclusions about stress under real time pressure.

Another limitation of this study is how the error rate was measured. Errors were identified by manually reviewing screen recordings, which may introduce some bias. Even though clear error categories were used, this method may be less reliable than automated data collection.

Another limitation is that participants completed the study in their own home environments. Factors such as screen size or possible distractions could have affected the results. While this reflects real-life conditions, it also reduces the level of control in the experiment.

The use of ADHD medication is another factor that may have influenced the results. Since medication use was not controlled, differences in attention and cognitive performance between participants could have affected the findings.

There are several ways the study could be improved in the future. For example, using a larger and more diverse sample would strengthen the results. A more controlled testing environment could also increase reliability. In addition, using a fully functional system or providing realistic rewards might better simulate real-world stress. Automated error logging would also improve the accuracy of the measurements.

Overall, the study has relatively high internal validity because of its structured design. However, external validity is more limited due to the experimental setting. Reliability can be considered moderate, as consistent procedures were followed, but some factors such as manual error analysis and differences between participants may have affected consistency.

Despite these limitations, the methods used were sufficient to meet the purpose of the study and answer the research questions. The results provide useful insights into how microinteraction design can affect both performance and user experience for people with ADHD in time-pressured digital environments.

6. Conclusions and further research

This chapter presents the overall conclusions of the study and reflects on its contributions in relation to the research purpose and questions. It summarizes the key findings regarding how microinteraction design influences task performance and user experience in time-pressured digital environments for users with ADHD.

In addition, the chapter discusses the implications of these findings for both theory and practice, and outlines suggestions for future research based on the limitations and insights of the study.

6.1. Conclusions

The purpose of this study was to investigate how microinteraction design influences task performance, error rates, and perceived stress during time-pressured online ticket booking for users with ADHD. By comparing two prototypes one including microinteractions and one without this study aimed to provide a deeper understanding of how small interface elements impact both objective performance and subjective user experience in cognitively demanding digital environments.

The results show a clear and consistent pattern across both objective and subjective measures. The prototype that included microinteractions resulted in faster task completion times and significantly fewer errors. In addition, participants reported lower levels of perceived stress and cognitive load, as well as higher usability and a stronger sense of control. These findings indicate that microinteractions can play a crucial role in supporting users with ADHD in time-pressured digital situations.

Answer to Research Question 1 (RQ1):
Microinteraction design had a measurable positive effect on task performance. Participants completed the booking tasks more quickly and with fewer errors when interacting with the prototype that included real time feedback, progress indicators, and guidance. This suggests that microinteractions can improve both efficiency and accuracy in time-sensitive digital environments.

Answer to Research Question 2 (RQ2):
Microinteraction design also had a significant impact on user experience. Participants reported lower perceived stress and cognitive load, alongside higher usability and confidence, when interacting with the microinteraction supported prototype. These results indicate that microinteractions can reduce uncertainty and support users' ability to maintain focus under time pressure.

From a theoretical perspective, the findings support Cognitive Load Theory by demonstrating that well designed microinteractions can reduce extraneous cognitive load through clear feedback, guidance, and visual structure. For users with ADHD, who may experience limitations in working memory and attentional regulation, these design elements appear to function as external cognitive support. This enables more efficient information processing, reduces uncertainty, and helps maintain focus throughout the task.

At the same time, the findings highlight that microinteraction design must be carefully balanced. While most participants benefited from the added feedback and guidance, some experienced animations and moving elements as distracting. This suggests that microinteractions are not universally beneficial in all forms, and that overly dynamic or visually intensive elements may introduce additional cognitive load for certain users. Therefore, the effectiveness of microinteractions depends not only on their presence, but also on how they are designed and implemented.

This study contributes to the field of Human Computer Interaction in several ways. First, it extends existing research by examining microinteractions in a commercial, time pressured context rather than educational or assistive environments. Second, it provides empirical evidence of how microinteraction design affects users with ADHD, a group that remains underrepresented in HCI research. Third, by combining objective performance measures with subjective user experience data, the study offers a more holistic understanding of how interface design influences both efficiency and experience.

From a practical perspective, the findings suggest several design implications. Digital ticket booking systems and similar time sensitive platforms should incorporate immediate and clear feedback after user actions, visible progress indicators that guide users through each step, and proactive error prevention mechanisms. At the same time, designers should avoid excessive or unnecessary animations, or provide options for users to customize or reduce dynamic elements, in order to accommodate different cognitive preferences and sensitivities.

In conclusion, microinteractions have the potential to significantly improve both performance and user experience in time pressured digital environments. When carefully designed, they can reduce cognitive load, lower stress, and support users in completing complex tasks more efficiently. These findings highlight the importance of designing beyond a “one-size-fits-all” user model and instead considering cognitive diversity as a central aspect of interface design.

Future research should further investigate how different types of microinteractions influence users with varying cognitive profiles, as well as explore adaptive or personalized interface designs that can dynamically respond to individual user needs in real-world settings.

6.2. Further research

Future research could expand on the findings of this study by including a larger and more diverse participant sample. A small sample size limits the ability to generalize results, and future studies with more participants would allow for stronger statistical analysis and more reliable conclusions about the effects of microinteraction design for users with ADHD.

In addition, future research could investigate different types of microinteractions separately in order to better understand which specific design elements contribute most to reduced cognitive load and improved performance. For example, animations, feedback systems, and progress indicators could be tested independently rather than as a combined design solution.

Another relevant direction for future research is to study microinteraction design in more realistic, high-stakes environments. In this study, the booking task was simulated without real consequences, which may have reduced stress levels. Future studies could therefore include real bookings or incentive-based tasks to better reflect authentic time pressure and decision-making conditions.

An additional direction would be to shift the design perspective by developing prototypes specifically tailored to users with ADHD. Instead of comparing a standard interface with a microinteraction-enhanced version, future studies could design interfaces based on accessibility principles from the outset, taking into account ADHD-related cognitive characteristics such as reduced working memory capacity and challenges with sustained attention. This approach would allow researchers to investigate how design can proactively support neurodivergent users, rather than evaluating accessibility as an added feature.

It would also be valuable to explore adaptive or personalized interface designs that adjust microinteractions based on user needs or cognitive profiles. Since the results indicate that some users experienced certain animations as distracting, customizable interfaces could be a promising direction for improving accessibility and usability for neurodivergent users.

Finally, future research could extend beyond users with ADHD and include other neurodivergent groups to better understand how different cognitive profiles interact

with microinteraction design in digital systems. This would contribute to a more inclusive understanding of usability and support the development of more universally accessible interfaces.

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8. Appendixes

Appendix 1 – Questionnaire

The questionnaire used in this study was implemented using Google Forms and is available at the following link:

https://docs.google.com/forms/d/e/1FAIpQLSdLRATPiKESmabmci04HAQc3zslCEJNpzAAZPh-wY228201Lg/viewform

This questionnaire was used after each completed task (Prototype A and Prototype B) to measure participants' subjective experience of the interaction. All items were answered immediately after task completion.

Participants rated their responses using a 7-point Likert scale, where:

1 = Strongly disagree

7 = Strongly agree

Section 1: Perceived Stress

- I felt a high level of stress during the task.
- Time pressure made it difficult for me to focus during the task.
- I felt calm and in control while completing the task. *(reverse coded)*

Section 2: Cognitive Load

- The task felt mentally demanding.
- I had to put a lot of effort into completing the task.
- The interface was easy to understand. *(reverse coded)*

Section 3: Usability and Interaction Experience

- I felt a strong sense of control while using the interface.
- The system provided clear feedback on my actions.
- I felt confident that my actions were correct.
- The visual elements helped me understand what to do next.
- The interface was easy to use overall.

Section 4: Qualitative Feedback

- What did you experience as most stressful during the task?
- What helped you stay focused during the task?
- Was there anything that felt confusing or distracting?

Section 5: Background Information (Context Variable)

- Were you under the influence of prescribed ADHD medication during the task?
  - Yes
  - No
  - Prefer not to say

Appendix 2 – Test Payment Card Information

A fictitious test credit card was used during the experiment to simulate a realistic payment process without involving real financial data.

Test Payment Card Information

The following identity and payment information was provided to all participants to ensure a standardized and secure testing environment:

- **Name:** Emma Lindqvist
- **E-mail:** emma.lindqvist.dev@gmail.com
- **Card Number:** 7639 7639 7510 9354
- Valid Thru: 10/30
- **CVC:** 456

Note: This data is fictional and was used strictly to measure task performance and input accuracy during the study.